

Chapter 2

The Log-Quantile-Density Model of the Volatility Smile

A volatility smile is a probability law written in Black coordinates. The log-quantile-density model builds that law first. Rank supplies unit probability; an exponential transport speed supplies a positive density; one shift supplies the forward. Two tail exponents cover the range from exponential to Gaussian log-return tails. One upper-share ledger then gives prices, digitals, density, local smile derivatives, moments, calibration sensitivities, and calendar order. A jointly calibrated maturity stack is constrained on the full real line, so its total-variance curves cannot cross.

2.1 A smile is a probability law

A listed board contains finitely many option prices. A desk needs a price at every strike. The completed curve is differentiated to obtain digitals and densities, integrated to price variance contracts, and compared with other expiries. Smooth interpolation is not enough: the curve must remain the price curve of a probability law.

2.1.1 Normalize first

Fix an expiry T . Let F be its forward and D its discount factor. Under the T -forward measure set

$$X = \log \frac{S_T}{F}, \quad Y = e^X = \frac{S_T}{F}, \quad \mathbb{E}[Y] = 1. \quad (2.1)$$

The last equality is the forward, or martingale, condition. For a strike K write

$$k = \log(K/F), \quad y = e^k = K/F.$$

The normalized call and put prices are

$$c(k) = \mathbb{E}[(Y - e^k)^+], \quad P(k) = \mathbb{E}[(e^k - Y)^+], \quad c(k) - P(k) = 1 - e^k. \quad (2.2)$$

Multiplying by DF returns the cash price. Rates, dividends, and the currency unit have disappeared from the one-expiry problem.

The market quotes implied volatility rather than normalized price. If $w(k) = \sigma(k)^2 \tau$ is total implied variance, where τ is variance time, the normalized Black call is

$$B(k, w) = \Phi(d_+) - e^k \Phi(d_-), \quad d_{\pm} = -\frac{k}{\sqrt{w}} \pm \frac{\sqrt{w}}{2}. \quad (2.3)$$

The implied variance $w(k)$ is the unique positive solution of $B(k, w(k)) = c(k)$ [2]. Thus price and implied volatility are two coordinates for the same object.

2.1.2 The call curve and the law

Use normalized strike y , not log-strike k , for the next calculation. If Y has density f_Y , differentiation under the expectation gives

$$\frac{\partial c}{\partial y} = -\mathbb{P}(Y > y), \quad \frac{\partial^2 c}{\partial y^2} = f_Y(y) \geq 0. \quad (2.4)$$

This is the Breeden–Litzenberger identity [3]. The first derivative is minus a digital probability. The second is the terminal density.

The same statement has a traded reading. For a strike spacing δ , the symmetric butterfly

$$c(y - \delta) - 2c(y) + c(y + \delta)$$

is a portfolio with a nonnegative payoff. Dividing by δ^2 and letting δ shrink gives $f_Y(y)$. A negative second difference is therefore a negative price for a nonnegative payoff.

Definition 2.1 (Arbitrage-free slice). An arbitrage-free slice is a call curve generated by a positive random variable Y with $\mathbb{E}[Y] = 1$. If Y has no atom at zero and has a density, the curve is equivalently a decreasing convex function of $y > 0$ satisfying

$$(1 - y)^+ \leq c(y) \leq 1, \quad -1 \leq c'(y) \leq 0, \quad c(y) \rightarrow 0 \quad (y \rightarrow \infty),$$

with $c'(y) \rightarrow -1$ as $y \downarrow 0$.

Every condition has an immediate source. The upper bound follows from $(Y - y)^+ \leq Y$; the lower bound is Jensen’s inequality; the slope is a survival probability; and the right boundary follows from integrability of Y . Conversely, the second derivative of such a curve is a probability measure with mean one. A valid call curve and a mean-one law are therefore equivalent descriptions.

Convexity belongs to y , not to k . The chain rule gives

$$\frac{\partial^2 c}{\partial k^2} = e^k \{e^k f_Y(e^k) - \mathbb{P}(Y > e^k)\},$$

which is negative far in the left wing for an ordinary density. Testing convexity in log-strike rejects correct curves.

In volatility coordinates the same condition is less direct. Twice differentiating $c(k) = B(k, w(k))$ yields

$$f_X(k) = g_D(k) \frac{\varphi(d_-(k, w(k)))}{\sqrt{w(k)}}, \quad g_D(k) = \left(1 - \frac{kw'}{2w}\right)^2 - \frac{(w')^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2}. \quad (2.5)$$

A twice differentiable smile is butterfly-free exactly when $g_D(k) \geq 0$ for every k [8]. The LQD model avoids this nonlinear differential constraint by constructing f_X first.

2.1.3 Four axes and one calculation

The chapter uses four horizontal coordinates:

coordinate	meaning	main use
$y = K/F$	normalized strike	convexity and static arbitrage
$k = \log y$	log-moneyness	market smiles
$u \in (0, 1)$	percentile rank	probability and quantiles
$z = \log\{u/(1 - u)\}$	log-odds	integration and numerics

The logit map $u \mapsto z$ is a bijection from $(0, 1)$ to $\mathbb{R}$. Its inverse and density are

$$u = \Lambda(z) = \frac{1}{1 + e^{-z}}, \quad \rho(z) = \Lambda(z)\Lambda(-z). \quad (2.6)$$

The median is $z = 0$; the tenth and ninetieth percentiles are $z = \mp \log 9$. Moving one unit in z multiplies the odds by e .

The calculation developed below has a fixed order:

$$\text{rank } u \longrightarrow \text{log-odds } z \longrightarrow \text{return } x(z) \longrightarrow \text{law } f_X \longrightarrow \text{price } c \longrightarrow \text{smile } w.$$

Probability is fixed at the first arrow. Monotonicity is fixed at the second. The Black formula is used only at the last.

2.2 LQD coordinates

2.2.1 Quantile speed

Let $Q(u)$ be the quantile function of X . If U is uniform on $(0, 1)$, then $Q(U)$ has that law. Write $q(u) = Q'(u)$ for the quantile density. Differentiating $F_X(Q(u)) = u$ gives

$$f_X(Q(u)) = \frac{1}{q(u)}. \quad (2.7)$$

The function q is the speed at which return space is traversed as rank increases. Small q packs probability into a narrow return interval; large q spreads it out. A strictly positive q makes Q strictly increasing. Writing a formula for $\log q$ makes that positivity automatic.

An unbounded law requires q to diverge at $u = 0$ and $u = 1$. The usual LQD skeleton removes the factors u and $1 - u$. To cover both exponential and Gaussian log-return tails, introduce the positive logarithmic gauges

$$\ell_-(u) = 1 - \log u, \quad \ell_+(u) = 1 - \log(1 - u). \quad (2.8)$$

Choose tail exponents $\alpha_-, \alpha_+ \in [0, 1/2]$ and set

$$\log q(u) = -\log u - \log(1 - u) - \alpha_- \log \ell_-(u) - \alpha_+ \log \ell_+(u) + g(u). \quad (2.9)$$

Equivalently,

$$q(u) = \frac{e^{g(u)}}{u(1 - u)\ell_-(u)^{\alpha_-}\ell_+(u)^{\alpha_+}}. \quad (2.10)$$

This is the precise version of replacing the endpoint factor u by $u\{\log(1/u)\}^{\alpha_-}$, and similarly on the right. The expression $(\log u)^\alpha$ is avoided because $\log u < 0$ for $u \in (0, 1)$.

Because $du/dz = u(1 - u)$, the two leading endpoint singularities cancel:

$$\frac{dQ}{dz} = \frac{e^{g(\Lambda(z))}}{\ell_-(\Lambda(z))^{\alpha_-}\ell_+(\Lambda(z))^{\alpha_+}}. \quad (2.11)$$

Thus the distribution is obtained by integrating an ordinary positive function on the log-odds line.

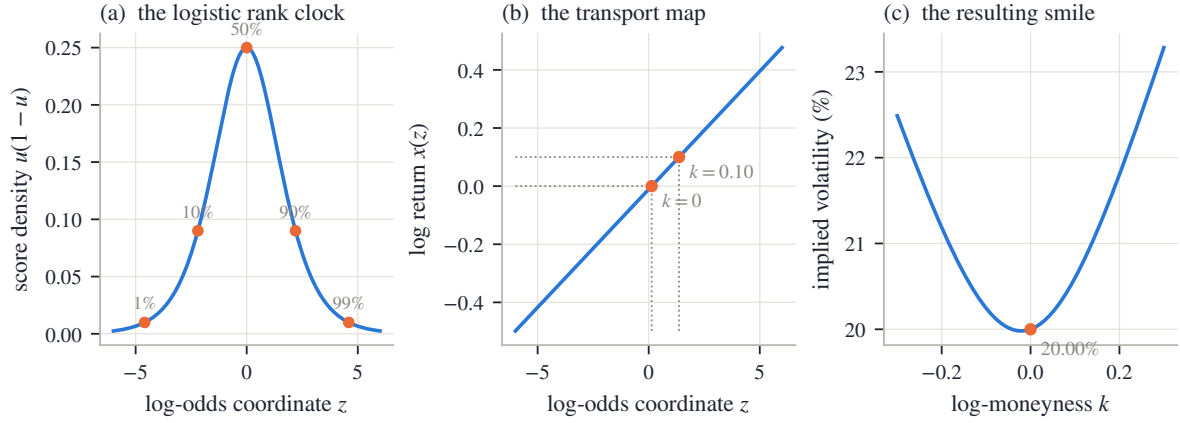


Figure 2.1: The transport construction in the exponential-tail subclass with constant speed. Left: logistic log-odds and five percentile marks. Center: an increasing map from log-odds to log-return; strike roots are read on this map. Right: the implied-volatility smile obtained by pricing the resulting law. The first two panels define the model; the last changes coordinates.

2.2.2 The generalized LQD slice

The body of the law is described by a low-degree polynomial in rank. Shifted Legendre polynomials are orthogonal under the uniform rank measure and remain well conditioned at moderate order.

Definition 2.2 (Generalized LQD slice). Fix $N \geq 2$, tail exponents $\alpha_{\pm} \in [0, 1/2]$, and coefficients $\theta = (L, R, a_2, \dots, a_N)$. Define

$$g(u) = (1-u)L + uR + \sum_{n=2}^N a_n P_n(1-2u), \quad (2.12)$$

$$x'(z) = \frac{e^{g(\Lambda(z))}}{\ell_-(\Lambda(z))^{\alpha_-} \ell_+(\Lambda(z))^{\alpha_+}}, \quad (2.13)$$

$$\boxed{X = x(Z), \quad x(z) = m + \int_0^z x'(t) dt, \quad Z \sim \Lambda.} \quad (2.14)$$

The shift m is chosen, when possible, so that $\mathbb{E}[e^X] = 1$.

For every finite coefficient vector, $x'(z) > 0$. Since $\alpha_{\pm} < 1$, its integral diverges at both ends, so x is a diffeomorphism from $\mathbb{R}$ to $\mathbb{R}$. Consequently

$$F_X(x(z)) = \Lambda(z), \quad f_X(x(z)) = \frac{\rho(z)}{x'(z)} > 0. \quad (2.15)$$

The map cannot fold, stop, or reverse. Positive density is a property of the coordinates, not a penalty imposed after calibration.

The endpoint coefficients are

$$\lambda_- = e^{g(0)} = \exp\left(L + \sum_{n \geq 2} a_n\right), \quad \lambda_+ = e^{g(1)} = \exp\left(R + \sum_{n \geq 2} (-1)^n a_n\right). \quad (2.16)$$

When $\alpha_{\pm} = 0$, they are the limiting speeds of the transport. For $\alpha_{\pm} > 0$, the limiting speeds are zero and $\lambda_{\pm}$ are the coefficients of their power-law decay in $|z|$. In both cases they are the natural tail-scale coordinates.

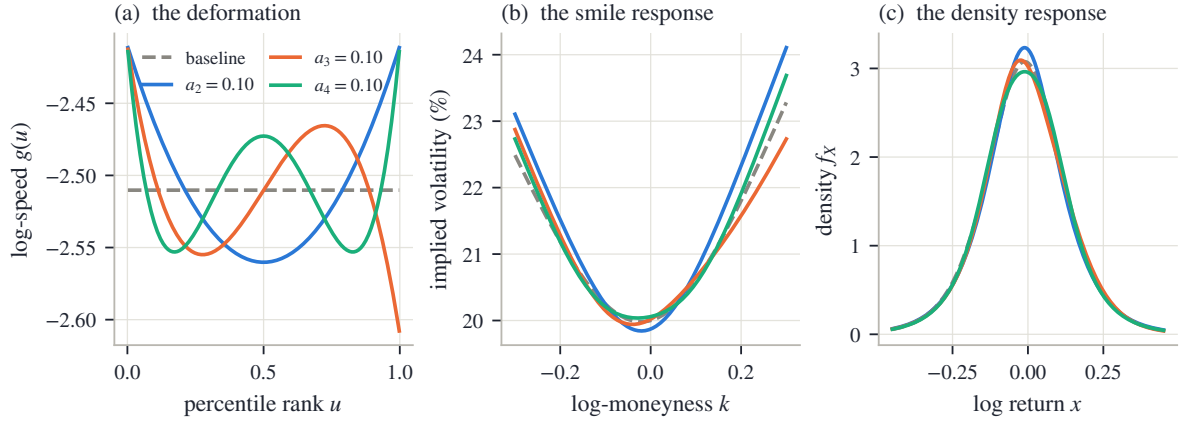


Figure 2.2: Three Legendre perturbations of one exponential-tail reference law. Left: their action on the log-speed g . Center: the implied-volatility response. Right: the density response. Raising g stretches return space and lowers density at the affected ranks. Even modes act on both sides; odd modes transfer shape from one side to the other.

Raw Legendre coefficients move both the body and the endpoints. For calibration it is preferable to use the endpoint-vanishing modes

$$\tilde{P}_n(u) = P_n(1 - 2u) - (1 - u) - (-1)^n u, \quad n \geq 2. \quad (2.17)$$

They satisfy $\tilde{P}_n(0) = \tilde{P}_n(1) = 0$. The equivalent parameter vector $(\log \lambda_-, \log \lambda_+, a_2, \dots, a_N)$ separates tail scales from body shape.

2.2.3 Martingale normalization

Let

$$\bar{x}(z) = \int_0^z x'(t) dt, \quad I = \int_{-\infty}^{\infty} e^{\bar{x}(z)} \rho(z) dz.$$

Proposition 2.3 (The forward condition). *If $I < \infty$, the unique martingale shift is $m = -\log I$. Moreover,*

$$I < \infty \iff \alpha_+ > 0 \text{ or } (\alpha_+ = 0 \text{ and } \lambda_+ < 1). \quad (2.18)$$

Proof. Since $X = m + \bar{x}(Z)$, $\mathbb{E}[e^X] = e^m I$, which gives the unique shift. If $\alpha_+ > 0$, then $\bar{x}(z) = O(z^{1-\alpha_+}) = o(z)$ while $\rho(z) \sim e^{-z}$, so the right integrand is integrable. If $\alpha_+ = 0$, then $\bar{x}(z) = \lambda_+ z + O(1)$ and the integrand is asymptotic to a constant times $e^{(\lambda_+-1)z}$. The left integrand is always integrable because $\bar{x}(z) \leq 0$ for $z \leq 0$ and $\rho(z) \sim e^z$. $\square$

The finite-forward wall $\lambda_+ < 1$ is therefore exact for the exponential-tail subclass and absent for every strictly lighter right tail. The change at $\alpha_+ = 0$ is real: an arbitrarily small positive exponent eventually dominates every linear exponential rate. It is also numerically nonuniform, a point used in section 2.6.

2.2.4 Two exact reference laws

Set $\alpha_{\pm} = 0$, $L = R = \log s$, and $a_n = 0$, with $0 < s < 1$. Then $x(z) = m + sz$ and X is logistic with scale s . Its moment-generating function is

$$\mathbb{E}[e^{tZ}] = \int_0^1 \left(\frac{u}{1-u} \right)^t du = \Gamma(1+t)\Gamma(1-t) = \frac{\pi t}{\sin(\pi t)}, \quad |t| < 1. \quad (2.19)$$

Hence

$$m = -\log \frac{\pi s}{\sin(\pi s)}, \quad \text{Var}(X) = \frac{\pi^2 s^2}{3}. \quad (2.20)$$

This is the primary end-to-end test of an implementation.

At the other endpoint of the tail range, a normal law $X \sim N(-s^2/2, s^2)$ has

$$q(u) \sim \frac{s}{u\sqrt{2\log(1/u)}} \quad (u \downarrow 0),$$

and the symmetric right relation. With $\alpha_{\pm} = 1/2$, the function g obtained from (2.9) has finite endpoint limits. Polynomial LQD log-speeds can therefore approximate the normal law without forcing g to diverge at either endpoint.

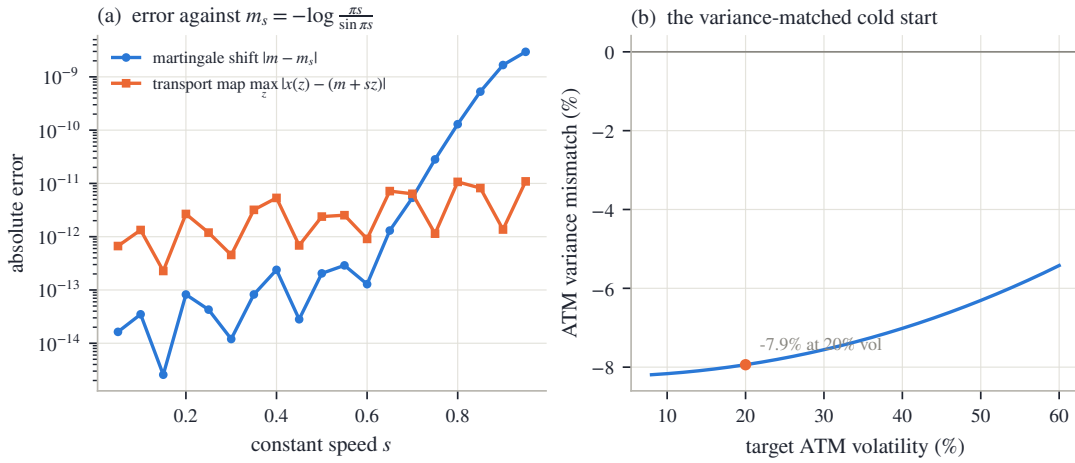


Figure 2.3: The constant-speed logistic slice as an implementation test. Left: computed transport and martingale shift errors against (2.20). Right: the variance-matched cold start and its small-variance ATM bias. The test exercises transport integration, normalization, tail treatment, and strike inversion with known answers.

2.3 Validity, tails, and smile wings

The LQD coordinates separate three facts. Positivity of density holds for every finite coefficient vector. The right tail determines whether the forward is finite. The two tail exponents determine how the implied smile continues beyond quoted strikes.

2.3.1 One-expiry no-arbitrage

Proposition 2.4 (Validity of one slice). *Let the parameters satisfy (2.18), and choose the shift of proposition 2.3. Then:*

- (i) X has a strictly positive continuous density on $\mathbb{R}$; $Y = e^X$ has a strictly positive density on $(0, \infty)$ and mean one;
- (ii) the call is strictly decreasing and strictly convex in normalized strike y , obeys the bounds in definition 2.1, and satisfies put–call parity;
- (iii) every nondegenerate finite-width butterfly has positive price;
- (iv) the implied-variance curve satisfies $g_D(k) > 0$ at every finite strike.

Proof. Equation (2.15) gives a positive continuous density and proposition 2.3 gives unit mean. In rank coordinates,

$$c(y) = \int_0^1 (e^{Q(u)} - y)^+ du.$$

Each integrand is decreasing and convex in y . Differentiation gives $c'(y) = -\mathbb{P}(Y > y)$ and $c''(y) = f_Y(y) > 0$. Jensen's inequality and dominated convergence give the boundary conditions. The butterfly and parity statements follow. Finally, (2.5) expresses the positive density as a positive Black factor times g_D . $\square$

This is a continuum result. It covers strikes between grid points and beyond the displayed smile. Numerical interpolation must preserve the same geometry; section 2.6.3 supplies the corresponding computation.

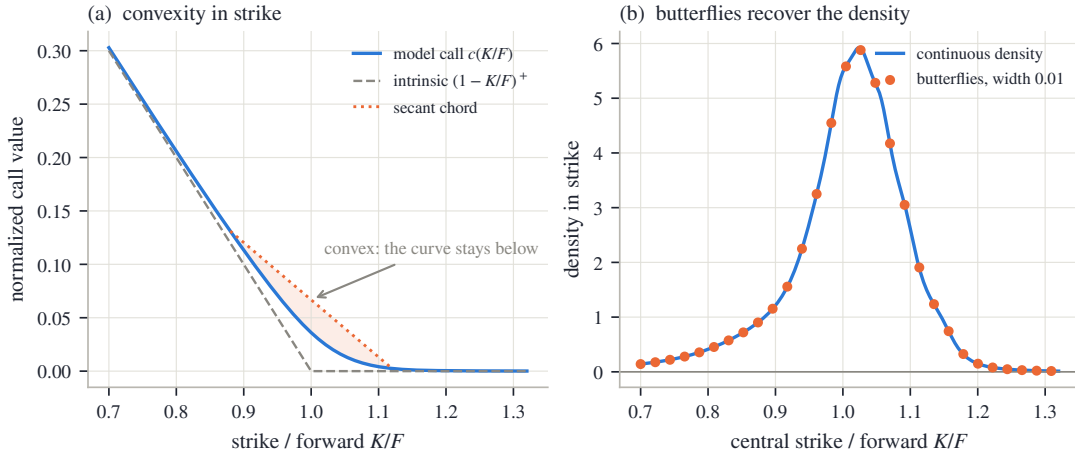


Figure 2.4: One-slice validity on the fitted SPY December 2026 law. Left: the call curve is convex in normalized strike. Right: symmetric butterfly prices, divided by squared half-width, converge to the density. The law supplies the convexity before calibration begins.

2.3.2 What the tail exponents mean

At the right endpoint, $1 - \Lambda(z) \sim e^{-z}$, $\ell_+(\Lambda(z)) = z + 1 + o(1)$, and $\ell_-(\Lambda(z)) = 1 + o(1)$. The left endpoint is symmetric. Integrating (2.13) gives

$$x(z) = m + b_+ + \frac{\lambda_+}{1 - \alpha_+} (z + 1)^{1 - \alpha_+} + o(1), \quad z \rightarrow +\infty, \quad (2.21)$$

$$x(z) = m + b_- - \frac{\lambda_-}{1 - \alpha_-} (1 - z)^{1 - \alpha_-} + o(1), \quad z \rightarrow -\infty, \quad (2.22)$$

for finite constants $b_{\pm}$. The additive constants do not affect the leading tail rates.

Proposition 2.5 (Log-return tail rates). *For a generalized LQD slice,*

$$-\log \mathbb{P}(X > x) \sim \left(\frac{(1 - \alpha_+)x}{\lambda_+} \right)^{1/(1 - \alpha_+)}, \quad x \rightarrow +\infty, \quad (2.23)$$

$$-\log \mathbb{P}(X < -x) \sim \left(\frac{(1 - \alpha_-)x}{\lambda_-} \right)^{1/(1 - \alpha_-)}, \quad x \rightarrow +\infty. \quad (2.24)$$

Proof. Invert (2.21) and use $\mathbb{P}(Z > z) = 1 - \Lambda(z) \sim e^{-z}$. The left proof is identical. $\square$

The two endpoints have the following exact interpretation.

- $\alpha_{\pm} = 0$: X has an exponential tail with scale $\lambda_{\pm}$; hence $Y = e^X$ has a power tail.
- $0 < \alpha_{\pm} < 1/2$: the log-tail is proportional to $-|x|^p$ with $1 < p < 2$.
- $\alpha_{\pm} = 1/2$: the log-tail is quadratic, as for a Gaussian. In this case a normal tail with standard deviation s corresponds to $\lambda = s/\sqrt{2}$.

The two sides may use different exponents. Equity smiles often retain $\alpha_- = 0$ on the crash side and use either 0 or a small positive value on the right. The choice is a tail convention unless far-wing instruments identify it.

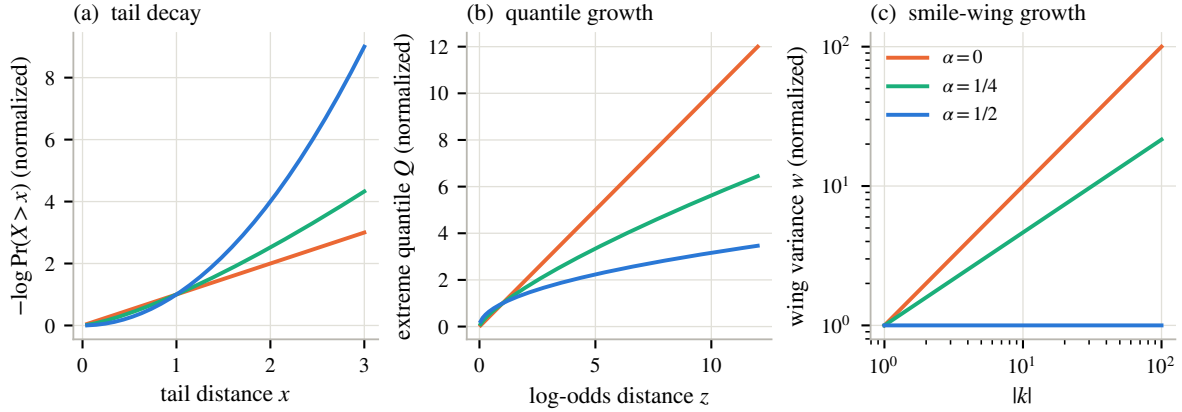


Figure 2.5: The tail exponent. Left: the leading negative log-tail grows like $x^{1/(1-\alpha)}$. Center: the extreme quantile grows like $z^{1-\alpha}$ in log-odds. Right: for $\alpha > 0$, total implied variance grows like $|k|^{(1-2\alpha)/(1-\alpha)}$; at $\alpha = 1/2$ it approaches a constant. Curves are normalized at one to compare rates, not levels.

2.3.3 Moments and the finite-forward wall

Define the two moment boundaries

$$\bar{r}_{\pm} = \begin{cases} 1/\lambda_{\pm}, & \alpha_{\pm} = 0, \\ +\infty, & \alpha_{\pm} > 0. \end{cases} \quad (2.25)$$

Proposition 2.6 (Moment domain). *For every generalized LQD slice,*

$$\mathbb{E}[e^{rX}] < \infty \iff -\bar{r}_- < r < \bar{r}_+. \quad (2.26)$$

Thus the last finite positive moment of Y beyond its mean is $r_+^ = \bar{r}_+ - 1$, and the last finite inverse moment is $r_-^* = \bar{r}_-$, with $+\infty$ allowed.*

Proof. Write the moment as

$$\mathbb{E}[e^{rX}] = \int_{-\infty}^{\infty} e^{rx(z)} \rho(z) dz.$$

If $\alpha_+ > 0$, then $x(z) = o(z)$ and the factor $\rho(z) \sim e^{-z}$ dominates every fixed positive r . If $\alpha_+ = 0$, the right integrand is asymptotic to a constant times $e^{(r\lambda_+-1)z}$. The left endpoint gives the other inequality in the same way. $\square$

The martingale condition is the point $r = 1$. When $\alpha_+ = 0$, it lies in the moment domain exactly when $\lambda_+ < 1$. A positive right tail exponent makes all positive moments finite. A positive left exponent makes all inverse moments finite.

2.3.4 From tail rate to smile wing

For exponential tails, Lee's moment formula gives the limiting slopes of total implied variance [12]:

$$\beta_R = \lim_{k \rightarrow +\infty} \frac{w(k)}{k}, \quad \beta_L = \lim_{k \rightarrow -\infty} \frac{w(k)}{|k|}. \quad (2.27)$$

Proposition 2.7 (Exponential-tail wing slopes). *If $\alpha_+ = 0$ and $\lambda_+ < 1$, and if $\alpha_- = 0$, then*

$$\beta_R = \frac{2\lambda_+}{(1 + \sqrt{1 - \lambda_+})^2}, \quad \beta_L = \frac{2\lambda_-}{(1 + \sqrt{1 + \lambda_-})^2}. \quad (2.28)$$

Both maps increase with the corresponding tail scale. For small λ , $\beta \sim \lambda/2$; as $\lambda_+ \uparrow 1$, $\beta_R \uparrow 2$.

Proof. Lee's function is $\Psi(r) = 2 - 4(\sqrt{r^2 + r} - r)$. Substitute $r_+^* = 1/\lambda_+ - 1$ and $r_-^* = 1/\lambda_-$ from proposition 2.6, then rationalize. The regular variation in proposition 2.5 turns Lee's limsup into a limit [1]. $\square$

Positive tail exponents have zero Lee slope because all moments are finite. Their sublinear wing growth remains explicit.

Proposition 2.8 (Intermediate and Gaussian-rate wings). *If $0 < \alpha_+ \leq 1/2$, then, as $k \rightarrow +\infty$,*

$$w(k) \sim \frac{1}{2} \left(\frac{\lambda_+}{1 - \alpha_+} \right)^{1/(1-\alpha_+)} k^{(1-2\alpha_+)/(1-\alpha_+)}. \quad (2.29)$$

If $0 < \alpha_- \leq 1/2$, the same formula holds on the left after replacing k by $|k|$ and the plus parameters by the minus parameters.

At $\alpha = 1/2$, total variance tends to $2\lambda^2$, the variance of the matching Gaussian tail. For $0 < \alpha < 1/2$, it diverges sublinearly. The limit $\alpha \downarrow 0$ is nonuniform: every positive α gives zero Lee slope, while $\alpha = 0$ gives the positive slope (2.28). A finite strike strip therefore cannot determine a small positive α reliably. Tail-exponent scenarios are more stable than an unconstrained per-slice estimate.

2.3.5 Coverage of the coordinates

For fixed $\alpha_{\pm}$, define the residual log-speed of a differentiable target quantile Q^* by rearranging (2.9):

$$g^*(u) = \log Q^{*'}(u) + \log u + \log(1 - u) + \alpha_- \log \ell_-(u) + \alpha_+ \log \ell_+(u). \quad (2.30)$$

Proposition 2.9 (Approximation at fixed tail class). *Suppose g^* extends continuously to $[0, 1]$ and the target law has mean one. If $\alpha_+ = 0$, also suppose $e^{g^*(1)} < 1$. Then polynomials of the form (2.12) can be chosen so that $g_N \rightarrow g^*$ uniformly. The corresponding quantiles converge on compact rank intervals, and their call curves converge uniformly over all strikes.*

The proof is in appendix 2.A. It is Weierstrass approximation followed by dominated convergence. The fixed tail skeleton supplies the common integrable envelope needed for martingale normalization and call prices.

No finite-order slice has an atom, a zero-density interval, bounded support, or mass at zero. Those features require a quantile jump, a flat part, or a finite endpoint, none of which is available to a positive smooth transport. Multimodality is allowed: the density rises where the transport slows and falls where it accelerates. Increasing basis order permits more such changes in the body without altering the stated tail class.

2.4 One ledger for prices and smile diagnostics

Once the transport x is known, one tail integral contains nearly every quantity required by a fitter.

2.4.1 The upper-share ledger

Definition 2.10 (Upper-share ledger). For an admissible slice define

$$G(z) = \int_z^\infty e^{x(t)} \rho(t) dt = \mathbb{E}[Y \mathbf{1}\{Z > z\}]. \quad (2.31)$$

The value $G(z)$ is the fraction of the unit forward carried by outcomes above rank $\Lambda(z)$. It satisfies

$$G(-\infty) = 1, \quad G(+\infty) = 0, \quad G'(z) = -e^{x(z)} \rho(z) < 0. \quad (2.32)$$

For a strike k , let z_k be the unique root of $x(z_k) = k$, and set $u_k = \Lambda(z_k) = F_X(k)$.

Proposition 2.11 (Call, put, digital, and density). *For every admissible slice,*

$$c(k) = G(z_k) - e^k(1 - u_k), \quad (2.33)$$

$$P(k) = e^k u_k - (1 - G(z_k)), \quad (2.34)$$

$$c'(k) = -e^k(1 - u_k), \quad (2.35)$$

$$c''(k) = -e^k(1 - u_k) + e^k \frac{\rho(z_k)}{x'(z_k)}, \quad (2.36)$$

$$f_X(k) = e^{-k} \{c''(k) - c'(k)\} = \frac{\rho(z_k)}{x'(z_k)}. \quad (2.37)$$

Proof. Exercise is the event $Z > z_k$. Splitting the payoff into its asset and cash legs gives

$$c(k) = \mathbb{E}[Y \mathbf{1}\{Z > z_k\}] - e^k \mathbb{P}(Z > z_k) = G(z_k) - e^k(1 - u_k).$$

The put follows on the complementary event. When the call is differentiated, the moving-root term is

$$\{G'(z_k) + e^k \rho(z_k)\} \frac{dz_k}{dk} = 0$$

because $G'(z_k) = -e^{x(z_k)} \rho(z_k) = -e^k \rho(z_k)$. This gives (2.35). Since $dz_k/dk = 1/x'(z_k)$, one more derivative gives (2.36) and (2.37). $\square$

Pricing is asset share minus strike cash. The digital probability is $1 - u_k$. The density is one division of two arrays already present in the transport. Neither digitals nor densities are obtained by differencing option prices.

Any integrable terminal payoff remains a rank integral:

$$\mathbb{E}[H(Y)] = \int_{-\infty}^{\infty} H(e^{x(z)}) \rho(z) dz. \quad (2.38)$$

In particular, the log-contract variance is

$$w_{\text{vs}} = -2\mathbb{E}[\log Y] = -2 \int_{-\infty}^{\infty} x(z) \rho(z) dz. \quad (2.39)$$

This is the law-side form of the usual option-strip replication [4, 13]. Under continuous paths and continuous monitoring it is the fair variance-swap strike. Jumps and discrete sampling require their standard corrections.

2.4.2 A worked option

In the constant-speed reference slice, choose $s = 0.081250$, which gives a six-month ATM volatility of twenty percent after exact repricing. The shift is $m = -0.010883$. For $k = 0.10$, the strike rank is $u_k = 79.65\%$. The ledger and cash legs are

$$G(z_k) = 0.247221, \quad e^k(1 - u_k) = 0.224876.$$

Therefore

$$c(0.10) = 0.247221 - 0.224876 = 0.022345,$$

which corresponds to 20.60% implied volatility. Every market option is evaluated by the same three operations: find the strike rank, read the ledger, subtract the cash leg.

2.4.3 The ATM microscope

Let z_* solve $x(z_*) = 0$ and define

$$c_0 = G(z_*) - (1 - u_*), \quad u_* = \Lambda(z_*), \quad f_* = \frac{\rho(z_*)}{x'(z_*)}. \quad (2.40)$$

The ATM Black variance and half-standard-deviation are

$$w_0 = 4 \left[\Phi^{-1} \left(\frac{1 + c_0}{2} \right) \right]^2, \quad d = \frac{\sqrt{w_0}}{2}.$$

Define the digital gap

$$\Delta = u_* - \Phi(d). \quad (2.41)$$

Proposition 2.12 (Exact ATM handles). *The total-variance smile $B(k, w(k)) = c(k)$ satisfies*

$$w(0) = w_0, \quad w'(0) = \frac{2\sqrt{w_0}}{\varphi(d)} \Delta, \quad (2.42)$$

$$w''(0) = \frac{2\sqrt{w_0}}{\varphi(d)} f_* - 2 + \left(\frac{1}{2w_0} + \frac{1}{8} \right) w'(0)^2. \quad (2.43)$$

In volatility units, $\sigma(k) = \sqrt{w(k)/\tau}$,

$$\sigma(0) = \sqrt{w_0/\tau}, \quad \sigma'(0) = \frac{\Delta}{\varphi(d)\sqrt{\tau}}, \quad (2.44)$$

$$\sigma''(0) = \frac{1}{\sqrt{\tau}} \left[\frac{f_*}{\varphi(d)} - \frac{1}{\sqrt{w_0}} + \frac{\sqrt{w_0}}{4} \left(\frac{\Delta}{\varphi(d)} \right)^2 \right]. \quad (2.45)$$

The derivation is in section 2.A.2. Its market reading is short. ATM price fixes level. The difference between the model digital and the matched flat-Black digital fixes skew. Density at the forward adds curvature. For a matched Black law, $u_* = \Phi(d)$ and the skew is zero. A negative equity skew means $u_* < \Phi(d)$, so the model digital $1 - u_*$ is richer than the flat-Black digital.

2.4.4 The analytic calibration Jacobian

The call depends on a parameter through both G and the root z_k . The moving-root term cancels exactly.

Theorem 2.13 (Envelope cancellation). *For every coordinate θ_j ,*

$$\frac{\partial c(k; \theta)}{\partial \theta_j} = \frac{\partial G(z; \theta)}{\partial \theta_j} \Big|_{z=z_k(\theta)}. \quad (2.46)$$

Proof. Differentiate (2.33). The coefficient of $\partial z_k / \partial \theta_j$ is $G'(z_k) + e^k \rho(z_k)$, which is zero by (2.32) and $x(z_k) = k$. $\square$

Let $b_j(u) = \partial g(u) / \partial \theta_j$ and $\bar{x} = x - m$. For fixed tail exponents,

$$\frac{\partial \bar{x}(z)}{\partial \theta_j} = \int_0^z x'(t) b_j(\Lambda(t)) dt, \quad (2.47)$$

$$\frac{\partial m}{\partial \theta_j} = - \int_{-\infty}^{\infty} e^{x(z)} \frac{\partial \bar{x}(z)}{\partial \theta_j} \rho(z) dz, \quad (2.48)$$

$$\frac{\partial G(z)}{\partial \theta_j} = \int_z^{\infty} e^{x(t)} \left(\frac{\partial m}{\partial \theta_j} + \frac{\partial \bar{x}(t)}{\partial \theta_j} \right) \rho(t) dt. \quad (2.49)$$

The denominator in the derivative of m is one because the accepted slice is already martingale-normalized. In raw coordinates,

$$b_L(u) = 1 - u, \quad b_R(u) = u, \quad b_{a_n}(u) = P_n(1 - 2u).$$

The same pass supplies the variance-swap row:

$$\frac{\partial w_{vs}}{\partial \theta_j} = -2 \int_{-\infty}^{\infty} \left(\frac{\partial m}{\partial \theta_j} + \frac{\partial \bar{x}(z)}{\partial \theta_j} \right) \rho(z) dz. \quad (2.50)$$

If a tail exponent is estimated, it adds the columns

$$b_{\alpha_-}(u) = -\log \ell_-(u), \quad b_{\alpha_+}(u) = -\log \ell_+(u) \quad (2.51)$$

to the derivative of $\log x'$. These columns grow in the remote ranks and are weakly identified by a finite strike strip. The reference policy fixes $\alpha_{\pm}$ and reports scenario fits instead.

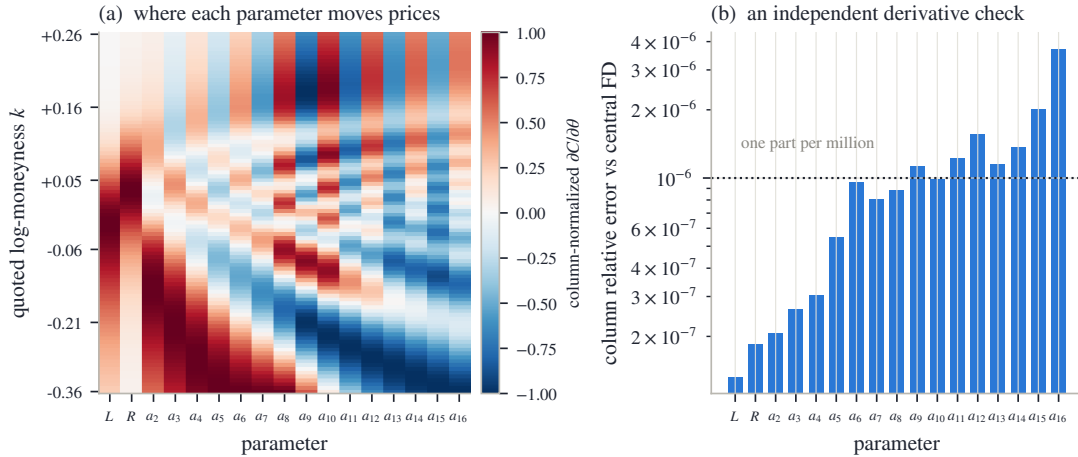


Figure 2.6: Analytic price sensitivities for the fitted SPY December 2026 slice. Left: one normalized sensitivity curve per parameter. Right: disagreement with an independent central difference. The comparison includes transport, normalization, ledger integration, root finding, and interpolation.

For $P = N + 1$ body and endpoint parameters and n_{grid} nodes, the sensitivity pass costs $O(Pn_{\text{grid}})$. It avoids rebuilding the complete slice and resolving every strike root for each finite-difference column.

2.5 Calibration

Validity is fixed by the coordinates. Calibration selects one law from the admissible family using the quoted strip, a basis order, a smoothness penalty, and a stated tail policy.

2.5.1 Fit prices in volatility units

Suppose the retained quotes are (k_i, σ_i) and $w_i = \sigma_i^2 \tau$. The basic one-slice objective is

$$\sum_i \left(\frac{c(k_i; \theta) - B(k_i, w_i)}{\partial_\sigma B(k_i, w_i) + \eta} \right)^2 + \omega_{\text{reg}} \sum_{n \geq 4} n^{2\nu} a_n^2. \quad (2.52)$$

Here $\partial_\sigma B = \varphi(d_+) \sqrt{\tau}$ is normalized Black vega and $\eta > 0$ is a vega floor. The residual is evaluated in price, so no implied-volatility inversion occurs inside the optimizer. Division by vega converts a central price error into the volatility units in which the market quotes. In a remote wing, the floor prevents a negligible option from receiving an unbounded weight.

The ridge starts at $n = 4$. The modes a_2 and a_3 carry the main curvature and asymmetry; higher modes describe progressively finer rank structure. The factor $n^{2\nu}$ makes that structure increasingly expensive.

An option quote is usually a bid–ask interval. If $[p_i^{\text{lo}}, p_i^{\text{hi}}]$ is the normalized price band and ϕ_i the vega normalizer, a band fit uses

$$\frac{(c_i - p_i^{\text{hi}})^+ + (p_i^{\text{lo}} - c_i)^+}{\phi_i}, \quad \sqrt{0.05} \frac{c_i - p_i^{\text{mid}}}{\phi_i}. \quad (2.53)$$

The first residual is zero inside the spread. The weak second residual selects one law when several laws fit a wide band equally well.

A volatility haircut h shrinks unreliable band edges toward the mid:

$$\sigma_i^{\text{lo}} = \min(\sigma_i^{\text{bid}} + h, \sigma_i^{\text{mid}}), \quad \sigma_i^{\text{hi}} = \max(\sigma_i^{\text{mid}}, \sigma_i^{\text{ask}} - h). \quad (2.54)$$

If the original spread is narrower than $2h$, the interval collapses to the mid. In the frozen sample this occurs for every featured SPY quote, while many wider NVDA bands remain active.

2.5.2 How many modes the strip can support

At order N the model has $N + 1$ body and endpoint parameters. A practical effective order is

$$N_{\text{eff}} = \max \left(\min \left(N, \left\lfloor \frac{n_q}{2} \right\rfloor - 1 \right), \min(N, 6) \right), \quad (2.55)$$

where n_q is the quote count. Away from the floor, this keeps roughly two quotes per shape degree of freedom. The rule is an identification convention, not a no-arbitrage condition. A larger order can change digitals and narrow butterflies between quotes while leaving vanilla residuals almost unchanged.

The quoted ranks make the same point directly. After a preliminary fit, record

$$[u_{\min}, u_{\max}] = \left[\min_i F_X(k_i), \max_i F_X(k_i) \right].$$

Modes acting mainly outside this interval are controlled by the ridge and the tail policy, not by quoted options. This rank interval should accompany the reported fit.

2.5.3 Optimization coordinates

Use the endpoint chart $(\log \lambda_-, \log \lambda_+, a_2, \dots, a_N)$ with the modes (2.17). Body moves then leave both tail coefficients fixed.

The right endpoint chart depends on the selected tail exponent:

$$\lambda_+ = \begin{cases} \expit \xi, & \alpha_+ = 0, \\ e^\xi, & \alpha_+ > 0. \end{cases} \quad (2.56)$$

The first branch makes the finite-forward wall unreachable. The second is unconstrained because every positive moment exists. A separate moment prior may still cap λ_+ for numerical or risk purposes.

The raw coefficients L and R follow from the endpoint equations (2.16). This conversion is linear once the body coefficients are fixed. Analytic derivatives in the endpoint chart are obtained by multiplying the raw-coordinate Jacobian by this small chart Jacobian.

2.5.4 Tail exponents are a surface policy

The exponent α changes the asymptotic class, not merely a local wing slope. At any finite strike range, a sufficiently small positive α can resemble $\alpha = 0$ while implying that every moment is finite and that the Lee slope is zero. Unconstrained per-slice estimation is therefore ill-conditioned.

The implementation uses one of three policies:

1. $\alpha_- = \alpha_+ = 0$ for exponential log-return tails and power gross- return tails;
2. fixed exponents chosen for an asset class or product, with separate left and right values allowed;
3. a small grid of exponent scenarios, each recalibrated, with downstream prices reported across the scenarios.

Within one underlier, the exponents are held common across expiry unless a monotone term-structure rule is imposed. This is required for stable full-line calendar order; section 2.7 gives the exact direction of the constraint.

2.5.5 Starting values and solve sequence

For $\alpha_{\pm} = 0$, the constant-speed seed

$$s = \frac{\sqrt{3w_0}}{\pi} \quad (2.57)$$

matches log-return variance. It is about eight percent low in ATM implied variance at small w_0 , so it is only a starting value. For other tail exponents, set all body modes to zero and solve one scalar equation for the common endpoint level that matches the ATM call. This costs one-dimensional root finding and avoids an arbitrary variance proxy.

A complete slice fit is:

1. infer F and D , convert strikes to k_i , and convert bid, mid, and ask volatilities to normalized prices once;
2. fix (α_-, α_+) , select N_{eff} , and construct the residual rows and weights;
3. initialize the endpoint chart from the scalar reference slice or from the preceding expiry;
4. at each step, build (x, x', G, G') on one grid, price every quote with (2.33), and obtain the Jacobian from (2.47)–(2.49);
5. rebuild the accepted parameters on the publication grid and compute ATM handles, moments, wing asymptotics, variance-swap strike, and numerical certificates from that object;
6. for several expiries, replace independent sequential fits by the joint calendar-constrained solve of section 2.7.3.

Market conventions are fixed before optimization. Every trial slice remains a law. All reported quantities are read from the final accepted law rather than from a separate smoothing pass.

2.6 Numerical realization

The mathematical model lives on the whole log-odds line. The implementation uses a finite central grid and analytic tail continuations. One build produces the transport, density ingredients, ledger, and all sensitivity arrays.

2.6.1 The central grid

Use a symmetric grid $[-Z_{\max}, Z_{\max}]$. At each node compute $u = \Lambda(z)$, the two gauges (2.8), $g(u)$, and $x'(z)$ from (2.13). Cumulative quadrature anchored at $z = 0$ gives $\bar{x} = x - m$. The martingale integral gives m . Reverse cumulative quadrature gives G .

The reference sizes are 2001 nodes during optimization and 8001 nodes for the accepted slice, with $Z_{\max} = 40$. Cubic Hermite interpolation uses the exact nodal derivatives

$$x'(z) > 0, \quad G'(z) = -e^{x(z)}\rho(z) < 0. \quad (2.58)$$

A bracketed Hermite root followed by one Newton step gives z_k .

Three expressions must remain in stable form. First,

$$\rho(z) = \text{expit}(z) \text{expit}(-z),$$

because $1 - \text{expit}(z)$ rounds to zero in the positive tail. Second, the cash leg is

$$e^k(1 - \Lambda(z_k)) = \exp\{k - \log(1 + e^{z_k})\},$$

evaluated with a log-sum-exp routine. Third, trial points are rejected before exponentiation if any grid value of g , x , or a sensitivity exponent exceeds the floating-point budget.

2.6.2 Tail continuation

For $z > Z_{\max}$, the leading transport continuation is

$$x_R(z) = x(Z_{\max}) + \frac{\lambda_+}{1 - \alpha_+} \left[(z + 1)^{1 - \alpha_+} - (Z_{\max} + 1)^{1 - \alpha_+} \right]. \quad (2.59)$$

The corresponding approximate root for $k > x(Z_{\max})$ is

$$z_R(k) + 1 = \left[(Z_{\max} + 1)^{1 - \alpha_+} + \frac{(1 - \alpha_+)(k - x(Z_{\max}))}{\lambda_+} \right]^{1/(1 - \alpha_+)}. \quad (2.60)$$

The formulas reduce to a straight line when $\alpha_+ = 0$. The left continuation is obtained by replacing $z + 1$ by $1 - z$ and reversing the sign of the transport increment.

In the exponential subclass, the leading missing martingale masses are

$$\int_{Z_{\max}}^{\infty} e^{\bar{x}} \rho \, dz \simeq \frac{e^{\bar{x}(Z_{\max}) - Z_{\max}}}{1 - \lambda_+}, \quad \int_{-\infty}^{-Z_{\max}} e^{\bar{x}} \rho \, dz \simeq \frac{e^{\bar{x}(-Z_{\max}) - Z_{\max}}}{1 + \lambda_-}. \quad (2.61)$$

For $\alpha_+ > 0$, the right correction is evaluated by one-dimensional log-domain quadrature under (2.59). Its endpoint approximation is

$$\int_{Z_{\max}}^{\infty} e^{x(z)} \rho(z) \, dz \simeq \frac{e^{x(Z_{\max}) - Z_{\max}}}{1 - x'(Z_{\max})}, \quad (2.62)$$

provided $x'(Z_{\max}) < 1$. The left denominator is $1 + x'(-Z_{\max})$.

The condition in (2.62) is computational, not probabilistic. If $\alpha_+ > 0$ and $x'(Z_{\max}) > 1$, the normalizing integrand continues to grow until the saddle $x'(z) = 1$ and then decays. For small

α_+ , that saddle can be extremely remote. A production implementation must either locate it and use scaled quadrature around it, or enforce the operational guard

$$x'(Z_{\max}) \leq 1 - \varepsilon_{\text{tail}}. \quad (2.63)$$

The reference implementation uses the guard. It excludes laws whose forward is mathematically finite but numerically dominated by ranks far beyond the grid.

For a strike beyond the right grid, evaluate the ledger under the same tail continuation and subtract the logistic cash leg. In the far limit,

$$c(k) \simeq e^{k-z_R(k)} \frac{x'(z_R(k))}{1 - x'(z_R(k))}. \quad (2.64)$$

At $\alpha_+ = 0$, this becomes $e^{k-z_R(k)}\lambda_+/(1 - \lambda_+)$. Deep left calls are evaluated as a small put plus parity, avoiding subtraction of two numbers close to one.

2.6.3 Numerical certificates

The continuous model is arbitrage-free; the stored interpolants must preserve that property. Four checks are applied to every accepted slice.

1. On every cubic Hermite segment for x , verify the Fritsch–Carlson box

$$0 \leq d_0 \leq 3s, \quad 0 \leq d_1 \leq 3s,$$

where s is the secant slope and d_0, d_1 are endpoint derivatives. Apply the same check to $-G$ [6].

2. Across the retained strike range, compute analytic w, w', w'' and insert them in (2.5). Require $g_D \geq -\varepsilon_{\text{dens}}$, where the tolerance is stated in units of the local Black density factor.
3. Rebuild on a grid with four times as many nodes. Compare prices, densities, ATM handles, and the martingale integral at off-grid strikes and ranks.
4. Test tail formulas separately in the exponential logistic reference law and in a Gaussian-rate reference obtained from $X \sim N(-s^2/2, s^2)$ with $\alpha_{\pm} = 1/2$.

The Hermite box is sufficient rather than necessary, but it has a useful margin: the smallest distance of (d_0, d_1) from its four faces. A small margin detects under-resolution before monotonicity is visibly lost.

The accepted parameter vector is rebuilt only once on the publication grid. Prices, densities, ATM handles, variance-swap values, and calendar ledgers all read that same immutable object. Quote residuals, basis order, tail policy, and numerical margins are reported separately because they measure different properties of the fit.

2.7 Calendar order on the whole line

One-expiry validity removes butterfly arbitrage. A surface must also order its expiries. The requirement is global because every fitted slice defines prices at every strike.

2.7.1 The calendar inequality

Lemma 2.14 (Equal-log-moneyness calendar order). *Assume deterministic rates and carry. Let $F_{0,t}$ be the forward curve and $M_t = S_t/F_{0,t}$ a positive martingale. For $T_1 < T_2$ and every $k \in \mathbb{R}$,*

$$c_1(k) \leq c_2(k). \quad (2.65)$$

Proof. With $Y_i = M_{T_i}$, conditional Jensen gives

$$\begin{aligned} c_2(k) &= \mathbb{E}[(M_{T_2} - e^k)^+] \\ &\geq \mathbb{E}[(\mathbb{E}[M_{T_2} \mid \mathcal{F}_{T_1}] - e^k)^+] = \mathbb{E}[(M_{T_1} - e^k)^+] = c_1(k). \end{aligned}$$

□

Equal k means strikes $K_i = F_i e^k$, not equal cash strikes. The two laws of Y_i have the same mean and increase in convex order. If carry is stochastic, the numeraire and the objects compared must be specified before using this relation.

Black call price is strictly increasing in w at every finite k . Hence

$$c_1(k) \leq c_2(k) \iff w_1(k) \leq w_2(k). \quad (2.66)$$

Full-line calendar order is therefore exactly the requirement that the stacked total-variance curves never cross.

2.7.2 Calls and ledgers contain the same order

Lemma 2.15 (Ledger–call conjugacy). *For every admissible slice,*

$$\begin{aligned} c(k) &= \max_{z \in [-\infty, \infty]} \{G(z) - e^k(1 - \Lambda(z))\}, \\ G(z) &= \min_{k \in [-\infty, \infty]} \{c(k) + e^k(1 - \Lambda(z))\}. \end{aligned} \quad (2.67)$$

The optimizers are $z = z_k$ and $k = x(z)$.

Proof. For fixed k ,

$$G(z) - e^k(1 - \Lambda(z)) = \int_z^\infty (e^{x(t)} - e^k) \rho(t) dt.$$

The integrand is negative before z_k and positive after it. Starting the integral at z_k maximizes the expression and gives the call. The first identity implies $c(k) + e^k(1 - \Lambda(z)) \geq G(z)$, with equality at $k = x(z)$, which gives the second identity. □

Theorem 2.16 (Calendar order is ledger order). *For two admissible mean-one slices,*

$$c_1(k) \leq c_2(k) \text{ for every } k \in \mathbb{R} \iff G_1(z) \leq G_2(z) \text{ for every } z \in \mathbb{R}. \quad (2.68)$$

Proof. If $G_1 \leq G_2$, take maxima in the first line of (2.67). If $c_1 \leq c_2$, take minima in the second line. □

The ledger uses a common rank axis across expiries and is already stored by the slice builder. It is therefore the natural coordinate for both calendar constraints and their final certificate. Ordered marginals are compatible with at least one martingale by the theorems of Strassen and Kellerer [11, 16]. The marginal surface does not select that martingale dynamics.

2.7.3 Full-line constraints in calibration

A constraint restricted to the common quoted span produces an ordered quote interpolant, not an ordered surface. It is appropriate only when the model is undefined outside that span. An LQD slice is defined on all of $\mathbb{R}$, so the accepted maturity stack imposes

$$G_{j+1}(z) - G_j(z) \geq 0 \quad \text{for every } z \in \mathbb{R} \quad (2.69)$$

for every adjacent pair of expiries.

The expiries are fitted jointly. The data term is the sum of their slice objectives. The inequalities (2.69) are hard constraints, not finite-weight residuals. An interior-point or sequential quadratic-programming solver can use the analytic ledger sensitivities from (2.49). A least-squares implementation can use an active-set outer loop, but a surface is accepted only after the full-line certificate below passes.

Global control couples unquoted wings across maturity. This is necessary: the extrapolated wings belong to the published surface. The coupling is made stable by choosing the tail policy for the stack before fitting. With common tail exponents, eventual calendar order requires

$$\lambda_{-,j+1} \geq \lambda_{-,j}, \quad \lambda_{+,j+1} \geq \lambda_{+,j}, \quad (2.70)$$

apart from exact ties, where the next asymptotic constant decides the order. These inequalities are imposed in the endpoint chart. If exponents vary by expiry, the farther law cannot have a lighter asymptotic tail: on either side $\alpha_{\pm,j+1} > \alpha_{\pm,j}$ forces a calendar reversal sufficiently far out. Common $\alpha_{\pm}$ across a maturity stack is the simpler policy.

If the hard constraints are inconsistent with the quote bands, no law stack can satisfy both the selected tails and those bands. The solver reports the smallest quote-band relaxation needed for feasibility. Reducing the calendar domain would replace a global surface by a local interpolant and is not used.

2.7.4 A complete calendar certificate

Let

$$H(z) = G_2(z) - G_1(z).$$

At both endpoints H tends to zero. Its derivative has a simple sign:

$$H'(z) = \rho(z)\{e^{x_1(z)} - e^{x_2(z)}\}. \quad (2.71)$$

Thus every interior extremum of the ledger gap occurs at a crossing of the two quantile curves, $x_1(z) = x_2(z)$.

This gives a finite certificate for the stored surface:

1. isolate every root of $x_{j+1} - x_j$ on the central grid; for cubic Hermite transports this is a cubic root problem on each segment;
2. evaluate $G_{j+1} - G_j$ at those roots and at the two grid boundaries;
3. continue both transports with (2.59) and its left analogue, isolate any tail roots, and evaluate the ledger gap there;
4. check the limiting tail order from the exponents, endpoint coefficients, and, in a tie, the asymptotic constants.

Between consecutive roots the sign of H' is fixed, so H is monotone. Nonnegative values at all its minima prove (2.69) on the whole real line. No arbitrary strike cutoff is involved.

During calibration, the current violating minimum is added to the active constraint set. This exchange step is repeated until the certificate finds no negative minimum. In practice the active ranks are few because adjacent smooth quantile curves cross only a few times.

At extreme strikes, both Black price and vega can underflow while the implied variance remains finite. A publication chart therefore evaluates remote wings from (2.28) or (2.29), matched to the central inversion, rather than inverting numerically zero prices. The chart then displays the globally certified surface instead of numerical noise from an ill-conditioned inverse.

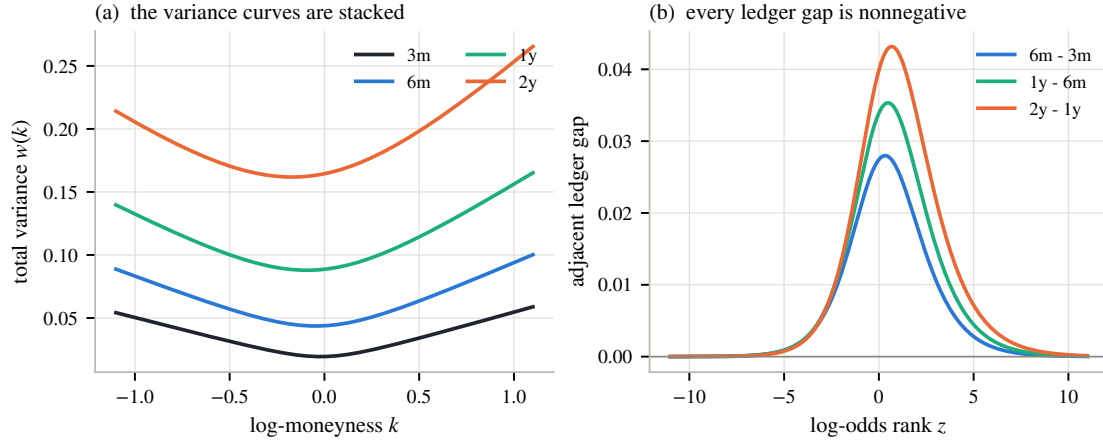


Figure 2.7: A globally ordered synthetic LQD maturity stack. Left: total variance at four expiries; the curves do not cross. Right: adjacent ledger gaps on the common rank axis; each remains nonnegative and returns to zero at both endpoints. The displayed curves illustrate the order; the quantile-root test certifies it on the full line.

2.8 Examples and operating contract

The frozen market examples use SPY and NVDA option chains from the 2026-08-03 US session, with eight expiries per underlier. They use the exponential-tail subclass $\alpha_- = \alpha_+ = 0$, order $N = 16$ subject to (2.55), haircut bands, and ridge $\omega_{\text{reg}} = 10^{-6}$ with $\nu = 1$. The examples therefore test the original LQD subclass; the generalized tail exponent is evaluated by separate scenario fits.

2.8.1 One liquid index slice

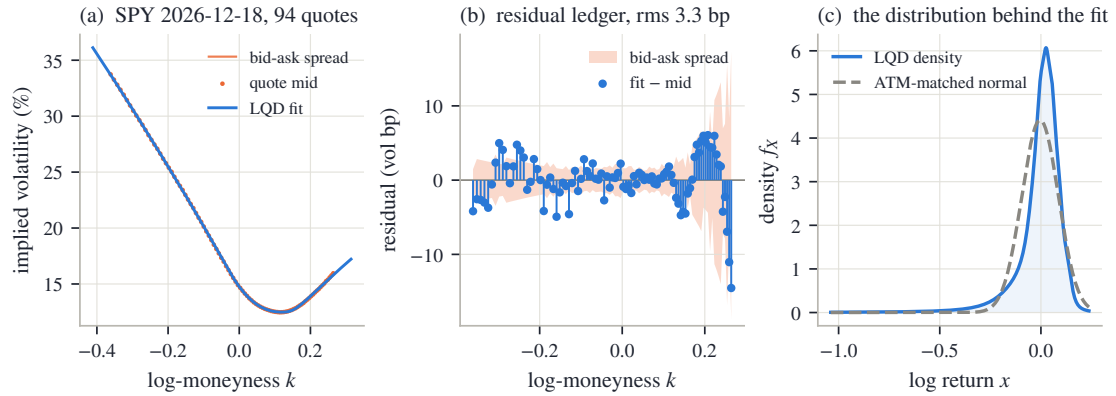


Figure 2.8: The SPY December 2026 slice. Left: quoted volatilities and the fitted smile. Center: signed residuals in volatility basis points. Right: the fitted density and an ATM-matched normal density. The left-shifted mass and richer model digital produce the negative equity skew.

The slice contains 94 quotes and fits them to 3.3 volatility basis points rms. The exact ATM formulas give volatility 14.76%, skew -0.430 , and curvature 4.40. The forward rank is 41.31%, below the matched Black rank, so $\Delta < 0$ and (2.44) gives a negative skew. The log-contract variance corresponds to 19.84% annualized volatility.

For the listed 800 call, $k = 0.0409$ and the strike rank is 65.27%. The asset-share and cash

entries are 0.378366 and 0.361834, hence

$$c(0.0409) = 0.378366 - 0.361834 = 0.016532.$$

This is \$12.70 per share and corresponds to 13.37% implied volatility.

The endpoint coefficients of the exponential subclass are $\lambda_- = 0.432$ and $\lambda_+ = 0.066$. They imply Lee slopes $\beta_L = 0.179$ and $\beta_R = 0.034$. These are asymptotic properties of the fitted continuation, not measurements at the outer quoted strikes.

2.8.2 A wider single-name strip

The long NVDA slice has endpoint coefficients $\lambda_- = 1.421$ and $\lambda_+ = 0.225$. Its left coefficient exceeds one. This is admissible because negative log-returns do not threaten the forward; it implies that $\mathbb{E}[Y^{-1}]$ is infinite in the exponential-tail subclass. A scenario with $\alpha_- > 0$ keeps the fitted central law but makes all inverse moments finite. Prices of inverse-sensitive or crash-tail positions distinguish the two policies more directly than vanilla fit error.

On the shortest NVDA expiry, 17 retained quotes give $N_{\text{eff}} = 7$. The order guard prevents a sparse strip from assigning an independent high-frequency shape mode to each quote.

2.8.3 What basis order changes

The synthetic benchmark is a martingale-normalized mixture of two lognormal components. Its density is bimodal, while its implied-volatility curve has only a mild shoulder.

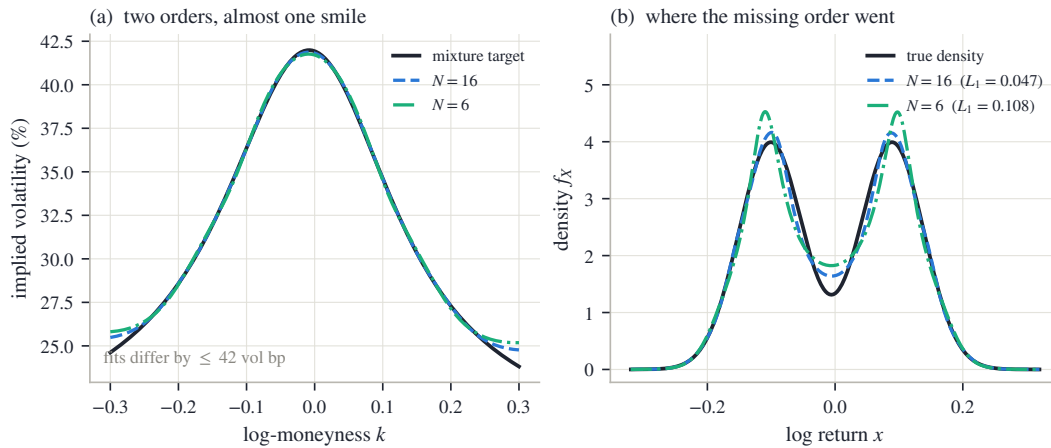


Figure 2.9: The same bimodal target fitted at orders 6 and 16. Left: the two implied-volatility curves nearly coincide over the quoted range. Right: the densities differ; the lower-order model shifts the modes and fills part of the trough. Vanilla residuals alone do not identify fine density structure.

At order 16 the L^1 density error is 0.047; at order 6 it is 0.108. Yet the implied-volatility curves differ by at most 41.8 volatility basis points on the comparison grid. Basis order should therefore be selected for the downstream use. Broad vanilla prices tolerate a lower order; digitals and narrow butterflies require a finer rank representation and stronger identification checks.

2.8.4 Relation to other smile models

Breeden–Litzenberger identify the call curve with a density [3]. Quantile modeling supplies the rank coordinate [9, 14]; Petersen–Müller use a log-quantile-density transform for density-valued data [15]; the metalog family is a related direct quantile expansion [10]. Lee and Benaim–Friz connect moments to implied-variance wings [1, 12].

SVI and SSVI specify total variance directly and characterize admissible parameter regions [7, 8]. Constrained price smoothers work directly with convex call curves [5]. LQD instead makes the law primary. Its main computational object is the pair (x, G) : the transport supplies density and the ledger supplies prices and calendar order.

2.8.5 The model contract

Derived	Fitted and checked	Chosen
Positive density, unit mass, unit forward, decreasing convex calls, and butterfly-free prices	Quote residuals, interpolation monotonicity, Durrleman margin, and grid convergence	Basis order, ridge, vega floor, quote-band rule, and numerical tolerances
Tail rate (2.23), moment domain (2.26), Lee slopes or sublinear wing law	Tail-quadrature accuracy, operational tail guard, and scenario sensitivity beyond quoted ranks	Tail exponents $\alpha_{\pm}$ and any additional endpoint-moment limits
Prices, digitals, density, ATM handles, log-contract variance, and analytic Jacobian from (x, G)	Multi-start stability, active calendar constraints, and the full-line quantile-root certificate	Common or monotone tail policy across expiry and the permitted quote-band relaxation
Full-line call order, ledger order, and noncrossing total variance are equivalent	The accepted maturity stack satisfies the order to its stated normalized-price tolerance	Dynamics between the fitted marginal laws are not selected by the surface

Table 2.1: The LQD operating contract. The first column follows from the model; the second is measured on the implementation; the third is supplied before calibration.

A fitted surface is published with its quote residuals, effective orders, quoted-rank spans, tail exponents and coefficients, moment limits, numerical margins, and minimum adjacent ledger gap. These quantities are sufficient to reconstruct how the surface is fitted, extrapolated, and certified.

2.A Proofs and asymptotic calculations

2.A.1 The volatility-space density factor

Let $c(k) = B(k, w(k))$. The identity $e^k \varphi(d_-) = \varphi(d_+)$ gives

$$\partial_k B = -e^k \Phi(d_-), \quad \partial_w B = \frac{\varphi(d_+)}{2\sqrt{w}}.$$

The required second partial derivatives are

$$\begin{aligned} \partial_{kk} B &= -e^k \Phi(d_-) + \frac{\varphi(d_+)}{\sqrt{w}}, \\ \partial_{kw} B &= \frac{d_+ \varphi(d_+)}{2w}, \\ \partial_{ww} B &= -\frac{d_+ \varphi(d_+)}{2\sqrt{w}} \partial_w d_+ - \frac{\varphi(d_+)}{4w^{3/2}}, \quad \partial_w d_+ = \frac{k}{2w^{3/2}} + \frac{1}{4\sqrt{w}}. \end{aligned}$$

Twice differentiating $c(k)$ and subtracting the first derivative gives

$$c'' - c' = (\partial_{kk} B - \partial_k B) + (2\partial_{kw} B - \partial_w B)w' + \partial_{ww} B(w')^2 + \partial_w B w''.$$

Now

$$2\partial_{kw} B - \partial_w B = -\frac{k\varphi(d_+)}{w^{3/2}}, \quad d_+ \partial_w d_+ = -\frac{k^2}{2w^2} + \frac{1}{8}.$$

Factoring $\varphi(d_+)/\sqrt{w}$ yields

$$c'' - c' = \frac{\varphi(d_+)}{\sqrt{w}} \left[\left(1 - \frac{kw'}{2w}\right)^2 - \frac{(w')^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2} \right].$$

Finally $f_X(k) = e^{-k}(c'' - c')$ and $e^{-k}\varphi(d_+) = \varphi(d_-)$, which gives (2.5).

2.A.2 The ATM identities

At $(k, w) = (0, w_0)$, with $d = \sqrt{w_0}/2$,

$$\begin{aligned} \partial_k B &= -(1 - \Phi(d)), & \partial_w B &= \frac{\varphi(d)}{2\sqrt{w_0}}, \\ \partial_{kk} B &= -(1 - \Phi(d)) + \frac{\varphi(d)}{\sqrt{w_0}}, & \partial_{kw} B &= \frac{\varphi(d)}{4\sqrt{w_0}}, \\ \partial_{ww} B &= -\frac{\varphi(d)}{4w_0^{3/2}} - \frac{\varphi(d)}{16\sqrt{w_0}}. \end{aligned} \tag{2.72}$$

From (2.35)–(2.37),

$$c'(0) = -(1 - u_*), \quad c''(0) = f_* - (1 - u_*).$$

One derivative of $B(k, w(k)) = c(k)$ gives

$$w'(0) = \frac{-(1 - u_*) + (1 - \Phi(d))}{\varphi(d)/(2\sqrt{w_0})} = \frac{2\sqrt{w_0}}{\varphi(d)} \Delta.$$

Two derivatives give

$$\partial_{kk} B + 2\partial_{kw} B w' + \partial_{ww} B (w')^2 + \partial_w B w'' = c''.$$

Substitution of (2.72) and the first-derivative formula reduces this to (2.43). Finally,

$$\sigma' = \frac{w'}{2\sqrt{\tau w}}, \quad \sigma'' = \frac{w''}{2\sqrt{\tau w}} - \frac{(w')^2}{4\sqrt{\tau} w^{3/2}},$$

which gives (2.44)–(2.45).

2.A.3 Sublinear wing asymptotics

Consider the right wing with $0 < \alpha_+ \leq 1/2$ and set

$$p = \frac{1}{1 - \alpha_+}, \quad A = \left(\frac{1 - \alpha_+}{\lambda_+} \right)^p.$$

Proposition 2.5 gives $-\log \mathbb{P}(X > x) \sim Ax^p$. The call has the tail-integral form

$$c(k) = \int_k^\infty e^x \mathbb{P}(X > x) dx.$$

Since $p > 1$, endpoint Laplace bounds give

$$-\log c(k) \sim Ak^p. \tag{2.73}$$

For example, for every $\varepsilon > 0$ and all large x , $e^{-(A+\varepsilon)x^p} \leq \mathbb{P}(X > x) \leq e^{-(A-\varepsilon)x^p}$; integrating the two bounds and then sending ε to zero proves (2.73).

All positive moments exist, so Lee's formula gives $w(k)/k \rightarrow 0$. Mills' ratio in the Black formula then gives

$$-\log B(k, w(k)) \sim \frac{k^2}{2w(k)}. \tag{2.74}$$

Equating (2.73) and (2.74) yields

$$w(k) \sim \frac{1}{2A} k^{2-p},$$

which is (2.29). The left proof uses the put and is identical.

2.A.4 Approximation at fixed tail class

Let g^* satisfy proposition 2.9. By Weierstrass' theorem there are polynomials g_N with $\varepsilon_N = \|g_N - g^*\|_\infty \rightarrow 0$. The endpoint chord and $P_2, \dots, P_N$ span every polynomial of degree at most N .

The speed ratio is bounded uniformly on the whole line:

$$e^{-\varepsilon_N} \leq \frac{x'_N(z)}{x^{*'}(z)} \leq e^{\varepsilon_N}.$$

Consequently $\bar{x}_N \rightarrow \bar{x}^*$ uniformly on every compact z interval. It remains to control the tails and the martingale shifts.

If $\alpha_+ > 0$, boundedness of the g_N gives constants C_1, C_2 such that

$$\bar{x}_N(z) \leq C_1 + C_2 z^{1-\alpha_+}, \quad z \geq 0.$$

Hence $e^{\bar{x}_N(z)}\rho(z)$ is dominated by an integrable multiple of $\exp(C_2 z^{1-\alpha_+} - z)$. If $\alpha_+ = 0$, the strict endpoint margin $e^{g^*(1)} < 1$ supplies $\bar{\lambda} < 1$ and a common bound $e^{\bar{x}_N(z)}\rho(z) \leq C e^{-(1-\bar{\lambda})z}$ for large z . On the left, $\bar{x}_N(z) \leq 0$ and $\rho(z) \leq e^z$. Dominated convergence therefore gives convergence of the normalizers and shifts.

The quantiles converge at every fixed rank. The same envelopes provide a common moment of order strictly larger than one, so the gross returns are uniformly integrable. Under the common uniform-rank coupling,

$$\sup_k |c_N(k) - c^*(k)| \leq \mathbb{E}|e^{Q_N(U)} - e^{Q^*(U)}| \rightarrow 0.$$

This proves uniform convergence of call prices.

2.A.5 Tail restrictions imposed by calendar order

On the right, (2.21) and endpoint integration give

$$\log G(z) = -z + \frac{\lambda_+}{1 - \alpha_+} z^{1-\alpha_+} + o(z^{1-\alpha_+}).$$

If a farther expiry has a larger α_+ , its positive correction has a smaller power and its ledger is eventually below the nearer ledger. If the exponents are equal, the larger λ_+ gives the larger eventual ledger. The left lower-share ledger satisfies

$$\log(1 - G(z)) = z - \frac{\lambda_-}{1 - \alpha_-} |z|^{1-\alpha_-} + o(|z|^{1-\alpha_-}), \quad z \rightarrow -\infty.$$

The same comparison gives the left conditions stated after (2.70). Equal exponents and equal endpoint coefficients leave the additive asymptotic constants to decide the order.

2.B Implementation blueprint

The listings record the numerical core. They omit only ordinary exception handling and the choice of quadrature library.

2.B.1 Stable grid quantities

The endpoint gauges should be evaluated from log-odds, not from rounded ranks:

$$\ell_-(\Lambda(z)) = 1 + \log(1 + e^{-z}), \quad \ell_+(\Lambda(z)) = 1 + \log(1 + e^z). \quad (2.75)$$

Both logarithms use a softplus routine. This remains accurate when $\Lambda(z)$ has rounded to zero or one.

Listing 2.1: Build one generalized LQD slice.

```

def build(theta, alpha, grid):
    # Endpoint chart: log lambda-, right-tail coordinate, body modes.
    lam_m, lam_p, a = decode_endpoint_chart(theta, alpha)
    L, R = raw_chord(lam_m, lam_p, a)

    u = expit(grid.z)
    ell_m = 1 + softplus(-grid.z)
    ell_p = 1 + softplus( grid.z)
    rho = expit(grid.z) * expit(-grid.z)

    g = (1-u)*L + u*R + a @ grid.Pmat
    log_dx = g - alpha.minus*log(ell_m) \
              - alpha.plus *log(ell_p)
    check_exponent_budget(log_dx)
    dx = exp(log_dx)

    if alpha.plus == 0 and lam_p >= 1 - wall_eps:
        raise InadmissibleForward
    if dx[-1] > 1 - tail_eps:
        raise RemoteSaddleBeyondGrid

    xbar = cumulative_simpson(dx, anchor=grid.mid)
    check_exponent_budget(xbar)

    I = integrate(exp(xbar)*rho)
    I += left_tail_mass(xbar[0], theta, alpha, grid)
    I += right_tail_mass(xbar[-1], theta, alpha, grid)
    m = -log(I)
    x = m + xbar

    dG = -exp(x)*rho
    G = reverse_integrate(-dG)
    G += right_tail_mass(x[-1], theta, alpha, grid)
    return Slice(x=x, dx=dx, G=G, dG=dG,
                 m=m, lam=(lam_m, lam_p),
                 alpha=alpha, theta=theta)

```

The two tail-mass functions use the continuation (2.59) and its left counterpart. For $\alpha = 0$ they use (2.61). For $\alpha > 0$ they integrate in log space after the change $z = Z_{\max} + t$. All tail corrections and their parameter derivatives enter both the primal build and the sensitivity pass.

The median anchor fixes $\bar{x}(0) = 0$. Simpson quadrature is fourth order on the uniform grid. Exact nodal derivatives make the cubic Hermite representation fourth order off grid without a global spline solve.

2.B.2 Call evaluation

Listing 2.2: Evaluate one normalized call.

```

def call_price(slc, k, grid):
    if k > slc.x[-1]:
        z = right_tail_root(slc, k, grid) # equation (2.54)
        share = right_ledger_tail(slc, z, grid)
        cash = exp(k - softplus(z))
        return share - cash

    if k < slc.x[0]:
        put = left_put_tail(slc, k, grid)
        return 1 - exp(k) + put

    z = bracketed_hermite_root(slc.x, slc.dx, k)

```

```

share = hermite_eval(slc.G, slc.dG, z)
cash = exp(k - softplus(z))
return share - cash

```

The right branch uses the same tail law as the martingale normalizer. A separate volatility extrapolation would generally break the density and calendar identities.

2.B.3 Sensitivity pass

Listing 2.3: All fixed-tail parameter sensitivities.

```

def sensitivities(slc, grid):
    # Rows are derivatives of g in the chosen coefficient chart.
    B = basis_columns_in_chart(slc.theta, grid.u)
    dxbar = cumulative_simpson(slc.dx * B,
                              anchor=grid.mid, axis=1)

    weight = exp(slc.x) * grid.rho
    dm = -integrate(weight * dxbar, axis=1)
    add_tail_derivatives_to_shift(dm, slc, grid)
    dx = dm[:, None] + dxbar

    integrand = weight[None, :] * dx
    dG = reverse_integrate(integrand, axis=1)
    add_tail_derivatives_to_ledger(dG, slc, grid)
    dG_slope = -integrand
    dvs = -2 * integrate(dx * grid.rho, axis=1)
    return Sensitivities(dG=dG, dG_slope=dG_slope, dvs=dvs)

def price_jacobian_row(slc, sens, k):
    z = strike_root(slc, k)
    return hermite_eval(sens.dG, sens.dG_slope, z)

```

The martingale derivative is $\partial m = -\int e^x \partial \bar{x} \rho dz$, including both tail pieces. Envelope cancellation then turns the ledger sensitivity at the strike root directly into the call sensitivity.

2.B.4 Full-line calendar constraints

During the joint maturity fit, maintain a small active set of ranks for every adjacent expiry pair.

Listing 2.4: Exchange algorithm for global calendar order.

```

def fit_surface(nodes):
    active = initial_calendar_ranks(nodes)
    while True:
        surface = constrained_joint_solve(nodes, active)
        worst = full_line_calendar_minima(surface)
        if worst.value >= -calendar_tol:
            return surface
        active[worst.pair].add(worst.rank)

def full_line_calendar_minima(surface):
    out = []
    for near, far in adjacent_pairs(surface):
        roots = all_pieewise_cubic_roots(far.x - near.x)
        roots += analytic_tail_roots(far, near)
        points = [-inf] + roots + [inf]
        gaps = [0.0] \
            + [far.ledger(z) - near.ledger(z) for z in roots] \
            + [0.0]
        out.append(min_with_location(gaps, points))
    return min(out)

```

If the difference of the two cubic transports vanishes on an entire segment, the ledger gap is constant there by (2.71); either endpoint represents the segment. Tail root isolation uses (2.59). The final certificate is rerun on the publication grid and with tighter tail quadrature than the optimizer uses.

2.B.5 Failure tests

The implementation treats the following states separately.

1. *Rank saturation*: use (2.75), two-sided logistic factors, and log-space cash legs.
2. *Interior overflow*: inspect the full arrays of g , $\log x'$, and $\bar{x}$ before exponentiation; admissible endpoints do not bound body modes.
3. *Exponential-tail wall*: for $\alpha_+ = 0$, keep $\lambda_+ \leq 1 - \varepsilon_{\text{wall}}$ in a logistic chart.
4. *Remote light-tail saddle*: for $\alpha_+ > 0$, apply (2.63) or use a saddle-centered quadrature.
5. *Near-zero option prices*: compare calls and calendars in log space and obtain plotted remote implied variance from the analytic wing law.
6. *Calendar gaps*: a finite penalty is not an acceptance test; the quantile-root certificate must pass for every adjacent expiry pair.

2.B.6 Reference defaults

Quantity	Default	Quantity	Default
Basis order N	16 (cap 24)	Order guard	(2.55)
Grid	$[-40, 40]$	Nodes	8001 (fit: 2001)
Tail exponents	stated scenarios	Exponential wall	$1 - 10^{-6}$
Remote-saddle margin	10^{-4}	Exponent budget	700
Ridge $(\omega_{\text{reg}}, \nu)$	$(10^{-6}, 1)$	Vega floor η	10^{-4}
Haircut h	0.005 volatility	Solver	constrained trust region
Slice check	801 strikes	Calendar check	all quantile crossings
Optimization tolerance	10^{-10}	Publication rebuild	fourfold check

Table 2.2: Reference numerical defaults. Tail exponents and their maturity policy are model inputs and are therefore reported with the fit.

The shortest porting sequence is: reproduce the logistic reference law; reproduce a normal law with $\alpha_{\pm} = 1/2$; implement the stable cash leg and both tail continuations; add the analytic sensitivity pass; then add the one-slice and full-line calendar certificates.

2.C Data and reproducibility

Every empirical number and inherited market figure in this alternative chapter comes from the frozen JSON snapshot used by the book. It contains SPY and NVDA option chains for eight expiries each from the 2026-08-03 US session, together with forwards, discount factors, retained quotes, fitted coefficients, and display curves. Reproduction requires no live market-data connection.

The stored fits use $\alpha_- = \alpha_+ = 0$. They are rebuilt from the frozen coefficients rather than refitted, so their prices and all macros remain those of the published snapshot. The two new figures in this version are synthetic: fig. 2.5 plots the derived asymptotic rates, and fig. 2.7 plots an exactly specified constant-speed LQD maturity stack. Neither introduces market observations.

Tail-policy comparisons recalibrate every exponent scenario under the same quotes and joint calendar constraints. Changing $\alpha_{\pm}$ at fixed body coefficients is only a local sensitivity.

Bibliography

- [1] Shalom Benaim and Peter Friz. Regular variation and smile asymptotics. *Mathematical Finance*, 19(1):1–12, 2009.
- [2] Fischer Black. The pricing of commodity contracts. *Journal of Financial Economics*, 3(1–2):167–179, 1976.
- [3] Douglas T. Breeden and Robert H. Litzenberger. Prices of state-contingent claims implicit in option prices. *Journal of Business*, 51(4):621–651, 1978.
- [4] Kresimir Demeterfi, Emanuel Derman, Michael Kamal, and Joseph Zou. A guide to volatility and variance swaps. *Journal of Derivatives*, 6(4):9–32, 1999.
- [5] Matthias R. Fengler. Arbitrage-free smoothing of the implied volatility surface. *Quantitative Finance*, 9(4):417–428, 2009.
- [6] Frederick N. Fritsch and Ralph E. Carlson. Monotone piecewise cubic interpolation. *SIAM Journal on Numerical Analysis*, 17(2):238–246, 1980.
- [7] Jim Gatheral. A parsimonious arbitrage-free implied volatility parameterization with application to the valuation of volatility derivatives. Presentation at Global Derivatives & Risk Management, Madrid, 2004.
- [8] Jim Gatheral and Antoine Jacquier. Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1):59–71, 2014.
- [9] Warren G. Gilchrist. *Statistical Modelling with Quantile Functions*. Chapman & Hall/CRC, Boca Raton, 2000.
- [10] Thomas W. Keelin. The metalog distributions. *Decision Analysis*, 13(4):243–277, 2016.
- [11] Hans G. Kellerer. Markov-Komposition und eine Anwendung auf Martingale. *Mathematische Annalen*, 198:99–122, 1972.
- [12] Roger W. Lee. The moment formula for implied volatility at extreme strikes. *Mathematical Finance*, 14(3):469–480, 2004.
- [13] Anthony Neuberger. The log contract. *Journal of Portfolio Management*, 20(2):74–80, 1994.
- [14] Emanuel Parzen. Nonparametric statistical data modeling. *Journal of the American Statistical Association*, 74(365):105–121, 1979.
- [15] Alexander Petersen and Hans-Georg Müller. Functional data analysis for density functions by transformation to a Hilbert space. *Annals of Statistics*, 44(1):183–218, 2016.
- [16] Volker Strassen. The existence of probability measures with given marginals. *Annals of Mathematical Statistics*, 36(2):423–439, 1965.